Future research roadmap: each study resolves a current limitation

Milestones are proposed research gates, not completed collaborations or validated findings.

CURRENT STYLIZED BENCHMARK VERIFIED COMPUTATION

Two operators, exogenous public congestion, independent binary confidentiality costs, M/P/F disclosure, pure BNE, no reporting error or implementation cost.

Current result: Medium congestion has the largest benchmark gap; CCAR improves the benchmark but also has no-effect, multiplicity, excessive-disclosure, and larger-gap regions.

STUDY 1 ECONOMICS

Question

Can a feasible rule induce truthful information quality and voluntary participation once implementation costs are included?

Method and assumptions

Bayesian mechanism design; participation and incentive constraints; regulator as a third governance actor.

Potential collaborator type

Mechanism-design or transport-economics researcher.

Success criterion and milestone

Characterize feasible regions; produce one counterexample and one implementable rule with explicit welfare and burden bounds.

STUDY 2 COMPUTER SCIENCE

Question

Does the mechanism remain safe and computationally useful with many heterogeneous drones, dynamic demand, and errors?

Algorithm and environment

Dynamic multi-agent simulation; privacy-preserving message exchange; robust optimization; stress and scalability tests.

Potential collaborator type

Multi-agent systems, air-traffic, or privacy engineer.

Success criterion and milestone

Predefine latency, constraint-violation, welfare-regret, and privacy metrics; release a reproducible 100-agent benchmark.

STUDY 3 BEHAVIORAL SCIENCE

Question

Do people understand the incentives, and do privacy salience, fairness, or reciprocity explain deviations from pure BNE?

Experimental design

Preregistered baseline-versus-CCAR choices with comprehension checks; students first, domain participants only after approval.

Ethics and collaborator type

IRB/ethics review as required; behavioral experimentalist.

Success criterion and milestone

Estimate choice and comprehension differences with uncertainty; do not generalize student play to drone operators.

POTENTIAL REAL-WORLD EVALUATION CONDITIONAL ON ALL PRIOR GATES

Regulator-supervised pilot with protected safety messaging, audit logs, appeals, calibrated implementation cost, and predeclared stop criteria.

Required evidence: operational reliability, safety, confidentiality burden, distributional access, and public accountability. None is established by the current simulation.

Han Zhang's three-party oversight suggestion enters Study 1 as a regulator/coordinator role; it remains proposed until formally modeled and tested.